

For each statement in 20-32, determine whether the statement is true or false. Prove the statement directly from the definitions if its true, and give counterexample if its false.

4.4.28. For all integers a , b and c , if $a|bc$ then $a|b$ or $a|c$.

Suppose a , b and $c \in \mathbb{Z}$.

We must show that if a is divisible by bc , then

a is divisible by b or a is divisible by c .

Argument form:

$\forall a, b, c \in \mathbb{Z}, a|bc \rightarrow a|b \vee a|c$

"For all integers a , b and c , if a divides bc ,

then a divides b or a divides c ."

Negation:

$\exists a, b, c \in \mathbb{Z}$ st $a|bc \wedge (a \nmid b) \wedge (a \nmid c)$

"There are integers a , b and c , such that

a divides bc , a does not divide b , and a does not divide c ."

Definition of divisibility:

" n is divisible by $d \leftrightarrow n = dk$, for some integer

k , and $d \neq 0$."

Let $a = 6$

Let $b = 3$

Let $c = 2$

$bc = 6$

$$a|bc = bc / a = 6 / 6 = 1 \in \mathbb{Z}$$

Therefore, $a|bc$.

$$a|b = b / a = 3 / 6 = 1 / 2 \notin \mathbb{Z}$$

$$a|c = 2 / 6 = 1 / 3 \notin \mathbb{Z}$$

Therefore, $a \nmid b$, and $a \nmid c$.

Solution: Since $a|bc$, but $a \nmid b$ and $a \nmid c$, the statement is false

by counterexample.